



Examination 2: Theoretical Part

1. Challenges in the Production of Knowledge

Auguste Rodin's sculpture *Gates of Hell* functions as an appropriate metaphor for the interdisciplinary nature of DH (Digital Humanities), as Terhi Nurmikko-Fuller (2023, p. 1) states. It is the blend of variously themed motifs within that sculpture that echoes the different domains in DH. Moreover, it depicts two areas that intersect within DH: the sciences and the arts. Given DH's broad domain, defining it is a mammoth task, akin to the term "interdisciplinary." Rather than arguing over which definition is best, Nurmikko-Fuller explains that it is more productive to view both concepts—DH and interdisciplinary—as a smoothie consisting of many elements,

mixed, modified, and rendered unrecognisable, that in the end create something requiring all of these subparts (pp. 1–3). This metaphor highlights how DH draws its strength from diversity, even when that diversity complicates attempts to define the field (cf. Nurmikko-Fuller, 2023, pp. 1–3).

Working in a field that interweaves various disciplines creates conflicts, Nurmikko-Fuller (2023) adds, due to differing opinions on how success is defined (p. 4). Conceptually, the sciences and the humanities are viewed as two distinct beasts, with the humanities portrayed as a field in which scholars work in isolation, while the sciences thrive on collaboration (p. 9). Scientific epistemology tends to emphasise objectivity, replicability, and the pursuit of stable truths, whereas humanistic epistemology values interpretation, context, and the coexistence of multiple possible meanings (Rochberg, 2007, pp. 1–3). Nevertheless, the humanities have influenced computational approaches from the beginning, with literary text analysis among the first DH projects. Furthermore, data used in the humanities played a role in the development of hypertext systems and served as a precursor to the Web (pp. 4, 9–12). These early contributions demonstrate that DH has always been shaped by cross-disciplinary exchange, even if this is not always acknowledged (cf. Nurmikko-Fuller, 2023, pp. 4, 9–12).

Nurmikko-Fuller (2023) describes the workflow of Linked Data (LOD) as involving many software tools, yet following a methodology for data publication that adheres to specific regulations and standards. It is measured by the Five-Star Standard, defined by Tim Berners-Lee, which requires the highest-quality data to be published using W3C standards and linked to other data online using RDF. RDF is a data model that represents an item using three properties, expressed as triples, for the concepts of subject, predicate, and object. In RDF, both the subject and the predicate are depicted as URIs, identifying the resources and the relationship between them. An object of a triple can be a URI, forming an interconnected network graph, with the subject and object as entities, while predicates connect these entities and give them meaning (pp. 13–14). This structure showcases how meaning is constructed in LOD, emphasising relationships rather than isolated facts (cf. Heath & Bizer, 2011, pp. 3–4).

Nurmikko-Fuller (2023) notes that LOD differs from web pages because HTTP URIs identify a resource and, when dereferenced, return a structured document describing that resource and its relationships to others, which are uniquely identifiable using URIs, URLs, and URNs. LOD assumes that entities are unambiguous and true, and that semantic web technologies such as OWL enable the representation of implicit or unproven statements. Further, the entities are considered intangible, viewed as concepts of an item or internationally agreed-upon identifiers of it (pp. 14–15). This abstraction is central to LOD's promise of interoperability, but it also introduces constraints (cf. Nurmikko-Fuller, 2023, pp. 14–15).

As real-world objects cannot themselves be returned via HTTP, servers use mechanisms such as 303 redirects to point clients to documents describing the entity (Heath & Bizer, 2011, p. 11). A key difficulty arises from distinguishing information resources from non-information resources. The W3C defines an information resource as one that has essential characteristics that can be fully conveyed in a single HTTP response (Sauermann & Cyganiak, 2008), such as a webpage or an image. Non-information resources—like people, places, or abstract concepts—cannot be returned directly; only descriptions can be provided. When a URI identifying a real-world entity is dereferenced, the server returns a document about the entity, creating ambiguity about whether the URI refers to the document or the thing itself. This ambiguity is a recurring challenge in LOD modelling and affects the reliability of entity queries (cf. Heath & Bizer, 2011, p. 11; Sauermann & Cyganiak, 2008, p. 2).

LOD struggles to represent entities whose details are uncertain or contested. For example, historical figures whose birth dates were never recorded or whose identities appear differently across sources cannot be represented without forcing a single, unambiguous value into the data model (cf. Nurmikko-Fuller, 2023, pp. 14–15). This limitation is also evident in Wikidata more broadly, where multiple possible Q-numbers or conflicting identifiers can exist for the same entity, yet LOD requires selecting a single identifier even when sources differ (cf.

Nurmikko-Fuller, 2023, pp. 15–17). Lastly, as mentioned before, the requirement for URIs to represent entities as fixed and unambiguous aligns with the sciences' preference for objectivity and observable data, whereas the humanities favour interpretation, contextual nuance, and openness to multiple possibilities, Rochberg (2007) notes, with both disciplines being pitted against each other in C. P. Snow's 1959 Rede Lecture (p. 1).

2. Confusion Regarding the Nature of Resources

Tom Heath and Christian Bizer (2011) state that LOD is a set of practices for publishing and interlinking structured web data, including using URIs as names for items, utilising HTTP URIs so that those names can be looked up, providing information using standards such as RDF or SPARQL, and including links to other URIs to enable discoverability. These principles support global sharing across servers and architectures and enable web crawlers to retrieve structured content from the web (p. 7). Together, these practices form the technical foundation that allows data from different providers to be combined into a single, coherent information space (cf. Heath & Bizer, 2011, pp. 3–4).

Building on these principles, Heath and Bizer (2011) explain that URIs identify not only web pages but also digital content, abstract ideas, and real-world entities such as people and places (p. 8). They refer to these as RDF links to distinguish them from ordinary hyperlinks. Furthermore, they state that a resource refers to anything identified by a URI, and that HTTP URIs provide globally unique names in a decentralised way, allowing domain owners to create identifiers (p. 9). While URLs and URIs follow the same specification, URLs identify the location of a resource, whereas URIs identify the resource itself. In contrast, URNs are unsuitable for LOD because they cannot be dereferenced, meaning they cannot return structured information when accessed (pp. 8–9). This distinction is central to LOD, which relies on the ability to retrieve resource descriptions directly from their identifiers (pp. 7–9).

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During the Wikidata exercise, the query intended to retrieve all operas composed by Mozart returned no results because the items representing Mozart's operas were not consistently typed as an instance of: opera (Q1344). The URIs existed, but the assigned resource type did not match the expected class, illustrating how LOD depends on correct classification of resources, and how confusion about the “nature” of a resource directly affects discoverability. Mistyped or inconsistently modelled resources distort research outcomes, making it difficult for scholars to rely on LOD for accurate retrieval: a concerning matter for DH research because inaccurate modelling directly affects the validity of queries, hindering researchers from asking the kinds of questions they want to answer and the conclusions they can draw (cf. Nurmikko-Fuller, 2023, pp. 14–17).

3. The Various Interests Involved

According to Dominic Oldman et al. (2016), LOD and the Semantic Web could be seen as representing two extremes, with institutions striving for innovations that could result in either something beautiful or chaotic, with humanistic scholars often feeling intimidated by LOD due to its scope and how it challenges traditional ideas embedded within that field. While seemingly approachable due to its user-friendly interface, it also compromises good research practices, such as transparency and reliability, by treating technology at face value rather than integrating it with accepted practices within the humanities (p. 252).

When reconciling data in OpenRefine, some cultural heritage datasets used local identifiers or institution-specific naming conventions that did not align with Wikidata's global identifiers. Smaller institutions often lack the resources or technical expertise to maintain LOD-ready vocabularies, while larger organisations can publish stable URIs and structured metadata. This highlights how differences in institutional capacity and expertise influence who can participate in LOD production and whose data becomes visible or remains isolated (Oldman et al., 2016, p. 253). Power imbalances, therefore, influence which knowledge becomes part of the global LOD ecosystem and which is rendered silent (cf. Oldman et al., 2016, pp. 252–253).

In LOD projects, various stakeholders must be considered, each bringing different priorities, constraints, and modelling assumptions. Paulheim (2017) explains that LOD ecosystems involve multiple actors, particularly data publishers and domain experts, whose modelling decisions and refinement strategies directly influence data quality and interoperability (pp. 492–495). Public-sector bodies may prioritise openness and accessibility, private-sector actors may emphasise efficiency or commercial value, and cultural institutions may focus on preservation and accuracy (cf. Paulheim, 2017, pp. 492–493). Community-driven platforms such as Wikidata rely on large, open contributor groups, while specialist datasets may be controlled by small expert communities (Paulheim, 2017, p. 494). These differing interests influence the standards, modelling practices, and compromises that determine how knowledge is represented and accessed on the Semantic Web (cf. Paulheim, 2017, pp. 492–495). As a result, LOD is not a neutral technical layer but a socio-technical system in which stakeholder priorities directly affect what becomes visible, how it is structured, and whose knowledge is privileged (cf. Oldman et al., 2016, p. 252; Paulheim, 2017, pp. 492–495).

(1497 words).

AI Policy

AI – Grammarly and suchlike – was used to streamline issues of repetition and tweak silly typos. It also helped me find a cat image.

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